

AI-Assisted Triage: Data Exploration

Link to Github notebooks: [Exploration and Visualization](#)

One-sentence verdict

The ED triage dataset appears good enough to build a triage machine learning model, but important limits must be handled carefully and the model must be validated locally before it is used to support real patient-care decisions.

Dataset summary

- Dataset: `yaleemmlc\_admissionprediction\_triage.csv`
- The dataset contains 55,121 emergency department arrival records and 225 columns.
- The main target variable is ESI (Emergency Severity Index), the triage level assigned to each patient. ESI runs from 1 (most urgent) to 5 (least urgent).
- The data includes demographics (age, gender, race, ethnicity), arrival and triage details, vital signs (heart rate, blood pressure, respiratory rate, oxygen saturation, temperature, glucose) and 200 binary chief complaint columns (each starting with cc_).
- Most patients fall in the middle acuity levels: ESI 3 is the largest group at 27,010 patients (49.0%), followed by ESI 2 at 17,924 patients (32.5%). ESI 1 is very rare with just 77 patients (0.1%), meaning there are few examples for a model to learn from when it comes to the sickest patients.
- There are 0 missing ESI labels, so every patient has a recorded triage outcome to train against.
- Overall, the dataset contains useful early clinical information, but the most urgent cases and many chief complaint features require careful handling before modelling.

Top 3 data-quality concerns

1. Very few of the most urgent (ESI 1) patients

Only 77 patients (0.1%) are ESI 1. This likely reflects the real ED case mix, but it creates a modelling concern because the algorithm has very few examples of the sickest patients to learn from. A model could look accurate overall while still missing ESI 1 patients. I would be cautious about creating synthetic ESI 1 rows because artificial examples may not accurately represent real high-risk patients. Planned response: In Week 6, report performance separately for each ESI group, especially ESI 1 and ESI 2, and consider class-weighting or careful feature grouping rather than inventing new patient records.

2. Many chief complaint features are too rare to use on their own

Of the 200 chief complaint columns, 149 appear in less than 0.5% of encounters. These rare columns are unreliable as standalone predictors since there isn't enough data behind them to trust the patterns. **Planned response:** Group, exclude, or treat rare chief complaint features cautiously during modelling.

3. Possible leakage columns

Columns such as 'disposition' and 'previousdispo' may reflect information that wasn't actually available at the time of triage. A real triage model can only use what a nurse would know at the time of assessment. **Planned response:** Exclude any columns that might give the model information it wouldn't actually have at triage time.

Top 3 reasons to proceed

1. There's a real triage outcome to learn from

With 0 missing ESI labels, the model can be trained on the actual triage decisions made, not a rough substitute for them.

2. The key triage information is included in the data

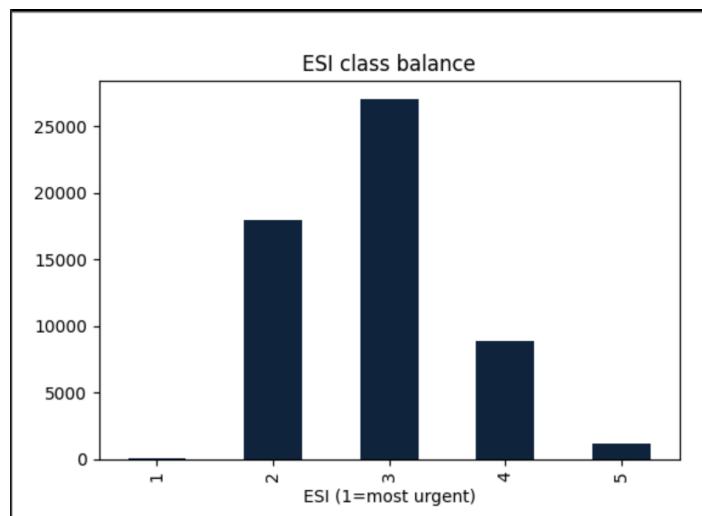
The dataset includes clinically meaningful information available early in a visit such as vital signs, demographics, arrival details and chief complaints.

3. Early patterns line up with clinical expectations

Some early findings match what a clinician would expect. For example, chest pain is associated with more urgent ESI levels and oxygen saturation showed one of the strongest vital-sign relationships with ESI, which makes clinical sense because low oxygen can be a sign of serious illness.

Data Visualizations and Key Findings

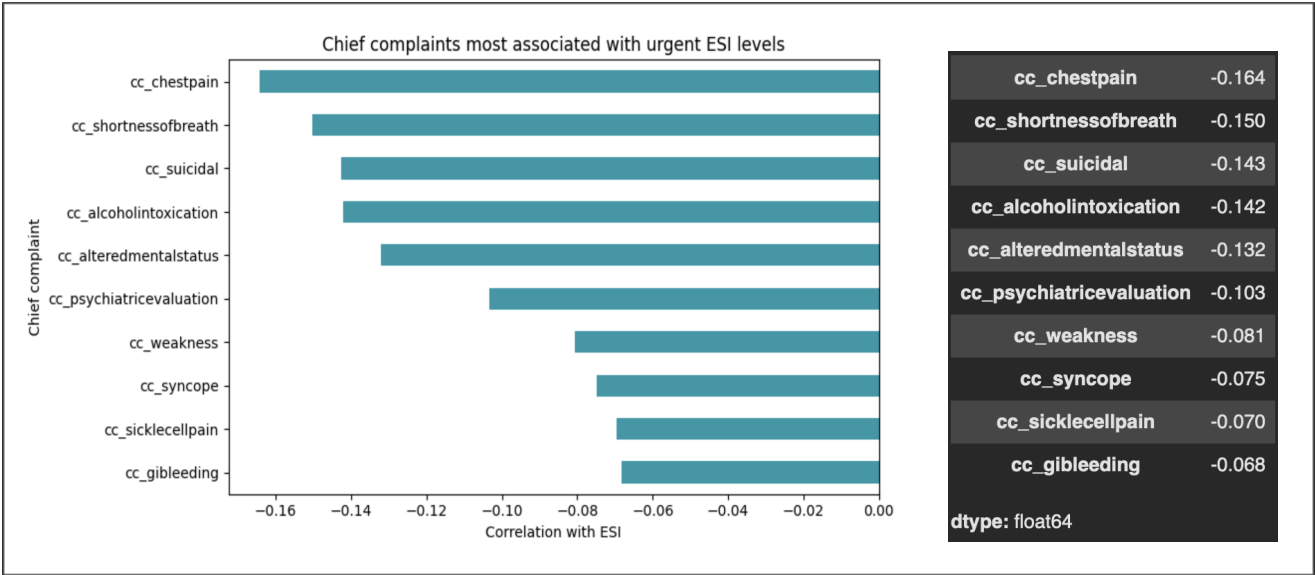
Figure 1: ESI class balance



1. ESI class balance

Figure 1 shows that most patients fall into ESI 2 and ESI 3, while ESI 1 cases are very rare. This supports the main concern that a model could perform well overall but still struggle to identify the most urgent patients.

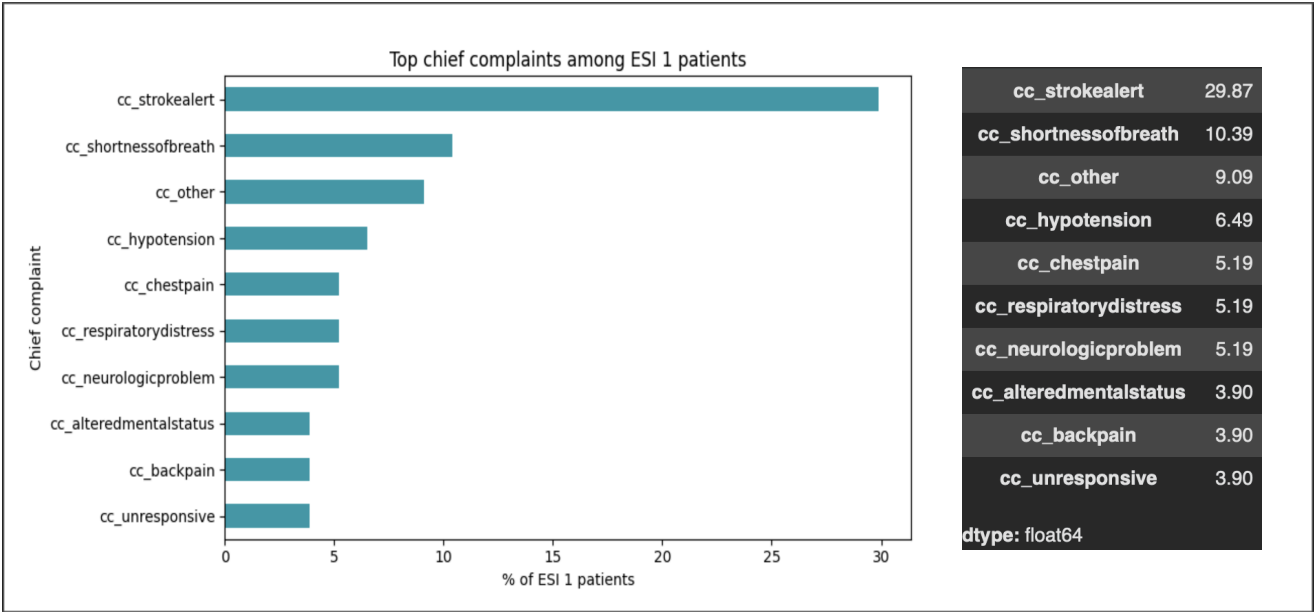
Figure 2: Chief complaints among ESI 1 patients



2. Chief complaints among ESI 1 patients

Figure 2 highlights the most common chief complaints among the sickest patients. These complaints are important to review because they may help guide which features should be tested for recognizing urgent cases in Week 6.

Figure 3: Chief complaints most associated with urgent ESI levels



3. Chief complaints most associated with urgent ESI levels

Figure 3 shows which chief complaints are most linked to urgent ESI scores. Because ESI 1 is the most urgent level, negative correlations suggest these complaints may be useful early warning features for the baseline model.

Caveats

- Correlation shows a relationship, not proof that a feature causes higher or lower acuity.
- The Week 6 model is a first attempt, not something ready for hospital use.
- Very few patients are the most urgent level (ESI 1), so performance on the sickest patients needs separate testing.
- Many chief complaint columns are rarely used and may need to be grouped before modelling.
- Temperature is recorded in Fahrenheit, so fever/low-temperature cutoffs should use Fahrenheit units, not Celsius.
- Disposition-related variables should be excluded to avoid leakage, so the model isn't trained on information it wouldn't actually have at triage time.
- The model should be tested in the actual setting where it will be used. One ED's data may not represent every Caribbean hospital or patient population.